Electrostatics

Preparation for the Balkan Physics Olympiad

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1 GENERAL ADVICE

Competition physics is a sport, not an art. Problems built around a single clever idea are dying out, and their place is being taken by Asian-style problems, where you have to work a calculator without looking at it. Training for a competition comes down to preparing for the specific test format you are aiming at, so it is very important to solve practice problems under a time limit. There is plenty of good general advice on Stefan Ivanov's website, so I see no point in repeating it here.

2 ELECTROSTATICS

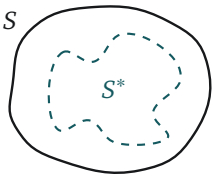
Fortunately, the number of ideas you need for electrostatics at Balkan Olympiad level is quite limited. Kevin Zhou's E1 and E2 handouts are more than enough to cover the topic completely. Harder problems often exploit the symmetry of the system.

2.1 *A little philosophy*

1 ●●●○ Squeezed foil

200 Puzzling Physics Problems

An isolated, bounded body whose surface S is made of metal foil has capacitance C. The foil is squeezed so that the new surface S^* lies entirely inside S (see the figure). Prove that the capacitance of the squeezed body is smaller than C.



2 ••○○ No stable equilibrium

David Tong

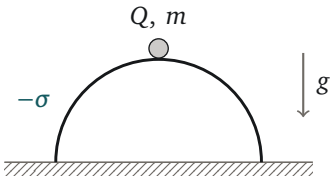
Prove that there is no stable electrostatic equilibrium: point charges in space cannot be arranged so that they are in stable equilibrium under electrostatic forces alone.[h2]

2.2 *Symmetry*

3 •••• A ball on a charged dome

Zhautykov Olympiad 2022

A thin dielectric hemispherical shell, uniformly charged with negative surface charge density $-\sigma$, rests on a horizontal table. A small ball of mass m is carefully placed on its top. Find the minimum charge Q of the ball for which the ball's position at the top of the shell is stable. The charges in the system do not redistribute (the shell stays uniformly charged, and the charges of the ball and the shell stay the same); the gravitational acceleration is g.



This problem needs some 3D geometry, as well as the integral

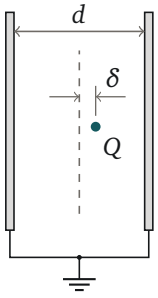
$$\int_{-\pi/2}^{\pi/2} \cos^2 x \, \mathrm{d}x = \frac{\pi}{2}.$$

4 ••●○ A charge between grounded plates

200 Puzzling Physics Problems

The plates of a parallel-plate capacitor are a distance d apart, and both are grounded. A point charge Q sits between the plates, a distance $\delta \ll d$ from the midplane (see the figure). What force acts on the charge? You may use

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} \approx 1.05.$$



HINTS

2. Use Gauss's law and argue by contradiction.